

Experiment one

Ohm's law

Aim

To determine the resistivity of two wires by plotting a graph for potential difference versus current

Apparatus

Two resistance wires, a voltmeter and an ammeter of appropriate range, a battery eliminator, a rheostat, a meter scale, a one-way key, connecting wires, and a piece of sandpaper

Theory.

According to Ohm's law, the current through a conductor is directly proportional to the applied voltage, provided temperature and other physical conditions of the conductor remain unchanged.

$$V \propto I$$

$$V = IR,$$

Where R is the resistance of the conductor.

Resistivity. $\rho = \frac{RA}{l}$ Where a is the area of cross-section and l is the length of the wire.

Procedure.

1. Arrange the apparatus in the same manner as given in the arrangement diagram. Determine the least count of the voltmeter and ammeter.
2. Insert the key K. Slide the rheostat. Contact and see that the ammeter and voltmeter are working properly.
3. Adjust the sliding contact of the rheostat such that a measurable current passes through the resistance coil or the resistance wire.
4. Note down the value of the potential difference V from the voltmeter and the current I from the ammeter.
5. Shift the rheostat contact slightly and take readings on both the ammeter and the voltmeter.
6. Take at least 5 sets of independent observations.
7. Plot a graph between V along the X-axis and I along the Y-axis. Find V/I by taking the slope between 2 points on the graph, and find the resistance of the first wire.
8. Note the cross-sectional area of the wire.
9. Find the resistivity of the first wire using the formula

10. Repeat the experiment for the second wire and find the resistivity of the second wire

Result.

Resistivity of wire 1=

Resistivity of Wire2 =

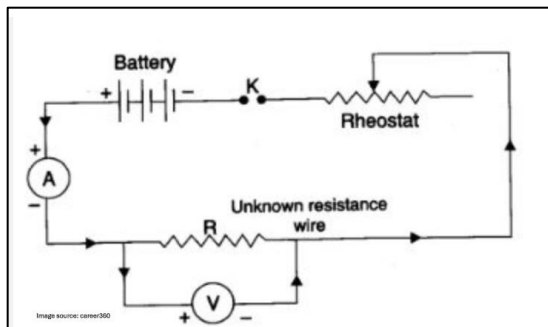
Precautions

1. The connections should be neat, clean, and tight.
2. The voltmeter and ammeter should be of the proper range.

Sources of Error

1. The instrument screws may be loose.
 2. Thick connecting wires may not be available.
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Circuit diagram on the left side of the first page and the observation table on the left side of the second page (with calculations)



Observations

Least count of the ammeter=

Least count of the voltmeter=

Wire 1

S.No	Voltmeter reading (V)	Ammeter reading (A)
1		
2		
3		
4		
5		

Wire 2

S.No	Voltmeter reading (V)	Ammeter reading (A)
1		
2		
3		
4		
5		

Screw Gauge readings- Area of cross-section of wires

L.C of screw gauge:

Zero error:

Wire 1

Main scale reading(MSR)	No of circular divisions / (CSD)	Circular scale reading (CSR=CSD xL.C)	Diameter (MSR+CSR)	Radius	Area of the wire 1

Wire 2

Main scale reading(MSR)	No of circular divisions / (CSD)	Circular scale reading (CSR=CSD xL.C)	Diameter (MSR+CSR)	Radius	Area of the wire 1

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Calculations:

Length of wire 1:

Resistance of wire 1 (from graph)=

Area of wire 1:

Resistivity of wire 1:

Length of wire 2

Area of wire 2:

Resistance of wire 2 (from graph)

Resistivity of wire 2

